

A study on the feasibility of using decision models to offload reasoning, and so lower inference cost

Reasoning models spend most of their tokens on questions they would have answered correctly anyway. A **decision model** — small, external, returning a calibrated probability instead of text — is asked per question whether the cheap answer is good enough, and reasoning is bought only when it changes the outcome: **3.1× fewer tokens** at **95%** of reasoning accuracy, or **21% fewer tokens** at no measurable accuracy loss.

Hasan Shabani · deepseek/deepseek-v4-flash-0731 · 660 held-out questions (AIME 2025, GPQA, MATH-500, MMLU-Pro, Stress suite)

3.1× fewer tokens than always reasoning 1,433 vs 4,465 per question	95% of reasoning accuracy kept 84.2% vs 88.5%	+4.6 pts above any equal-budget traffic split 95% CI +1.8 to +7.5	21% saved at no measurable accuracy cost -0.9 pts, McNemar p = 0.48
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COST-ACCURACY FRONTIER

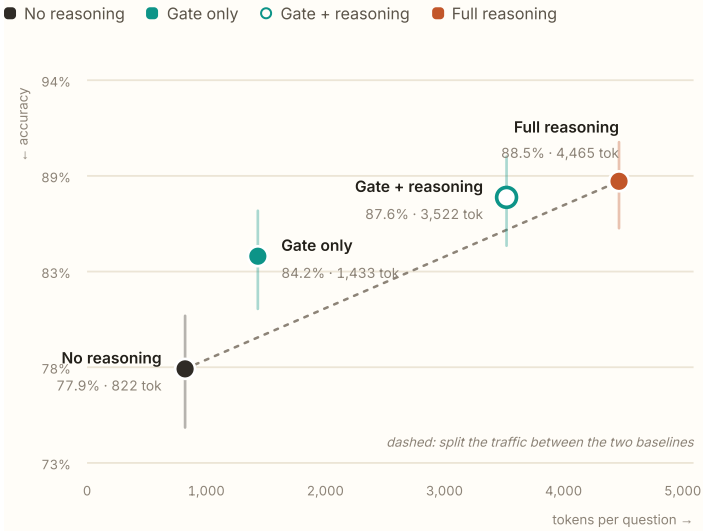
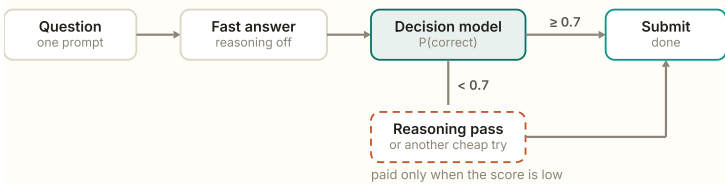


Figure 1. Tokens per question against accuracy; bars are 95% Wilson intervals. The dashed line is the honest baseline — the accuracy bought by routing a share of traffic to reasoning. Both gated policies sit above it; for *gate only* the gap is +4.6 points (95% CI +1.8 to +7.5, paired bootstrap).

MECHANISM



POOLED RESULTS

POLICY	ACCURACY	95% CI	TOKENS	VS REASONING	TOKENS / CORRECT
No reasoning	77.9%	75%–81%	822	-10.6 pts · 0.18×	1,056
Gate only	84.2%	81%–87%	1,433	-4.2 pts · 0.32×	1,701
Gate + reasoning	87.6%	85%–90%	3,522	-0.9 pts · 0.79×	4,021
Full reasoning	88.5%	86%–91%	4,465	reference	5,047

METHOD

- Answer cheaply** — the same model, reasoning disabled, one pass.
- Score it externally** — one yes/no question to a decision model, which returns a probability (1.27 calls per question, \$0.07 per thousand). Self-assessment does not separate right from wrong; an independent scorer does (AUC 0.88 against 0.70).
- Escalate only on doubt** — below 0.7, pay for the reasoning pass (*gate + reasoning*), or retry cheaply and keep the best-scored answer (*gate only*, which never reasons).

FINDINGS

Accuracy is preserved. Gate + reasoning is -0.9 points from always reasoning (22 wins, 28 losses, McNemar p = 0.48) at 21% fewer tokens.

Cheap is not enough on its own. Gate only beats one-shot answering by +6.4 points (p < 0.001) — 1,701 tokens per correct answer against 5,047 for reasoning.

It beats the obvious alternative. The gain over an equal-budget traffic split is +4.6 points, with the confidence interval clear of zero.

LIMITS

Every number here comes from a model pretrained and post-trained to reason on its own, with the gate strapped on afterwards — never taught to signal uncertainty or to lean on a checker. A model trained with a decision model in the loop should do better; this is a floor for the idea, not a ceiling.

The saving also tracks how much a model overthinks: on a local 35B model whose one-shot answers already matched its reasoning, the gate cut tokens but bought no accuracy. And gating every step of a multi-step agent episode hurts — an early veto derails a sound plan (100% solved by reasoning against 53% for the gate on 15 long-horizon tasks).